🔑 **Answer Key — Teacher Copy · Lesson 6-11**

MSTAR PRACTICE WORKSHEET

Lesson 6-11: Fraction Division Problem Solving

Grade 6 Mathematics · Standard 6.NOS.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A ribbon is 4/5 of a yard long. Each bow uses 1/10 of a yard. Which equation finds how many bows can be made?

A. 1/10 ÷ 4/5

B. 4/5 ÷ 1/10

C. 4/5 × 1/10

D. 4/5 + 1/10

The total (4/5 yard) is divided into groups of 1/10 yard each. The equation is 4/5 ÷ 1/10 = 4/5 × 10/1 = 40/5 = 8 bows.

Question 2 SELECTED RESPONSE — CORRECT: C

A detective has 1/2 gallon of solution. Each test uses 1/8 gallon. How many tests can be run?

A. 1/16

B. 1/4

C. 4

D. 16

1/2 ÷ 1/8 = 1/2 × 8/1 = 8/2 = 4 tests.

Question 3 SELECTED RESPONSE — CORRECT: B

A trail is 3/4 mile long. A jogger runs laps of 1/4 mile each. How many laps does the jogger complete?

A 1/3 lap

B 3 laps

C 3/16 lap

D 4 laps

3/4 ÷ 1/4 = 3/4 × 4/1 = 12/4 = 3 laps. This is reasonable: three 1/4-mile pieces fit into 3/4 mile.

Question 4 SELECTED RESPONSE — CORRECT: B

A board is 5/6 foot long. Each shelf bracket needs 1/6 foot. How many brackets fit?

A 1/36

B 5

C 5/36

D 6

5/6 ÷ 1/6 = 5/6 × 6/1 = 30/6 = 5 brackets.

Question 5 SELECTED RESPONSE — CORRECT: C

Marcus solved 2/3 ÷ 1/4 and got 8/3 = 2 2/3. He says 'That can't be right because I started with less than 1.' Is Marcus's math correct? Is his reasoning correct?

A His math and reasoning are both correct — the answer should be less than 1

B His math is wrong — the answer should be 1/6

C His math is correct (2 2/3), but his reasoning is wrong — dividing by a small fraction gives a larger result

D His math is wrong — the answer should be 8/12

2/3 ÷ 1/4 = 2/3 × 4/1 = 8/3 = 2 2/3. This IS correct. When you divide by a number less than 1, the quotient is greater than the dividend.

Question 6 SELECTED RESPONSE — CORRECT: C

Room 1 — Evidence Locker: Agent Torres has 7/8 pound of casting clay to lift footprints. Each cast uses 1/8 pound. How many casts can she make to unlock the door?

A 1/7

B 6

C 7

D 8

The total (7/8 lb) is the dividend and each cast (1/8 lb) is the divisor: 7/8 ÷ 1/8 = 7/8 × 8/1 = 56/8 = 7 casts.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

A ribbon is 4 1/2 feet long. Marta cuts the whole ribbon into pieces that are each 3/4 foot long. How many pieces does she get?

A. 1/6 of a piece

B. 3 3/8 pieces

C. 5 1/4 pieces

D. 6 pieces

The total amount 4 1/2 feet is the dividend and the size of one piece, 3/4 foot, is the divisor: 9/2 ÷ 3/4 = 9/2 × 4/3 = 36/6 = 6 pieces.

- **A:** That divides the piece size by the total. The whole ribbon is the dividend: 9/2 divided by 3/4.
- **B:** That multiplies rather than divides. Flip the divisor 3/4 to 4/3 first.
- **C:** Ask how many 3/4-foot pieces fit inside 4½ feet: 9/2 times 4/3.

Part B — correct answer: C

Which reasoning shows why your answer to Part A is correct?

A. Divide the size of each piece by the total length: 3/4 ÷ 9/2 = 3/4 × 2/9 = 6/36 = 1/6.

B. Add the total length and the size of each piece: 4 1/2 + 3/4 = 5 1/4.

C. The total amount is the dividend and the size of each group is the divisor, so total ÷ size = number of groups: 9/2 ÷ 3/4 = 9/2 × 4/3 = 36/6 = 6.

D. Multiply the total length by the size of each piece: 9/2 × 3/4 = 27/8 = 3 3/8.

Asking how many pieces fit means dividing the total by the group size, in that order. Reversing the order or choosing another operation changes what the answer counts.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, D

A jug holds 3/4 gallon of juice. Each cup holds 1/8 gallon. Select ALL statements that correctly describe this situation.

- **A** The dividend is 3/4 gallon, the total amount of juice in the jug.
- **B** The divisor is 1/8 gallon, the size of one group.
- **C** The equation that models the situation is 3/4 ÷ 1/8.
- **D** The jug fills 6 cups.
- **E** The answer is 3/32 gallon.
- **F** The answer must be less than 1 because both numbers in the problem are less than 1.

The total 3/4 gallon is the dividend and the cup size 1/8 gallon is the divisor, so the model is 3/4 ÷ 1/8 = 3/4 × 8/1 = 24/4 = 6 cups. The value 3/32 comes from multiplying instead of dividing, and dividing by a number smaller than the dividend gives a quotient larger than 1 here.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Write your own fraction division word problem where the answer is 8. Then show the equation, solve it step by step, and explain how you know your answer is reasonable.

Model answer: The safest way to build this problem is backwards: choose a group size, multiply it by 8, and let that product be the total in your story. For example, 8 groups of 1/2 make 4, so "4 cups of trail mix packed into 1/2-cup bags" gives 4 ÷ 1/2 = 4 × 2/1 = 8. The answer is reasonable because 8 groups of the size you chose rebuild the original total exactly.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.